CMUC Online Seminar on Algebra, Logic and Topology

University of Coimbra

Characterization of coextensive varieties of universal algebras

David Neal Broodryk

University of Cape Town, South Africa

10 November 2020 (Tuesday), 16:00 (Lisbon time) · Zoom

Abstract

A coextensive category can be defined as a category C with finite products such that for each pair X,Y of objects in C, the canonical functor $\times: X/C \times Y/C \to (X \times Y)/C$ is an equivalence. This is the dual of an extensive category, a term that was first used by W. F. Lawvere and S.Schanuel, although "categories with disjoint and universal coproducts" were considered by A. Grothendieck a long time ago, and there are related papers of various authors. A motivating example of a coextensive category is the category CRing of commutative rings, which leads one to ask which varieties of universal algebras are coextensive. In this talk we answer this question by giving a syntactical characterization of coextensive varieties of universal algebras.